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End Semester Examination May 2025

Name of the Course: B. Tech

Semester: VIII

Name of the Course: Earthquake resistant design of buildings

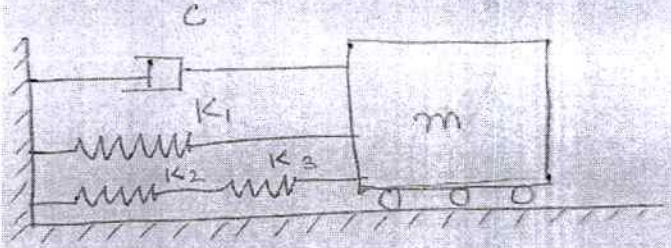
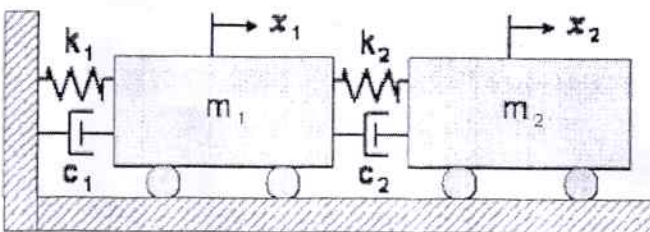
Course Code: TCE 801

Time: 3 hours

Maximum Marks: 100

Note:-

- (i) Answer all the questions by choosing any two of the sub questions.
- (ii) Each question carries 10 marks

Q1	(20marks)	
(a)	A vibrating system have mass 3 kg, stiffness 200 N/m and damping of 4Nsec/m. Find Zeta, natural frequency of damped vibration, logarithm decrement and ratio of two consecutive amplitudes?	CO1
(b)	A damper offers resistance 0.08 N at constant velocity 0.06 m/sec. damper is used with $k = 12$ N/m. Determine damping frequency of system when mass of system is 0.5 kg?	
(c)	Derive the equation for displacement of SDOF damped free vibration when structure is underdamped?	
Q2	(20 marks)	
(a)	Define the terms: frequency, mode of vibrations, fundamental frequency, normal mode of vibration and resonant frequency?	CO2
(b)	 For the given figure calculate the natural frequency and Damping ratio, if $m = 20\text{kg}$; $c = 100\text{kg/s}$; $K_1 = 3500\text{N/m}$; $K_2 = 250$ N/m and $K_3 = 1500$ N/m. Also state whether the system is overdamped, underdamped or critically damped.	
(c)	Write down the equation of motion for the given multi degree of freedom system?	
		
Q3	(20 marks)	
(a)	How surface waves are more disastrous than body wave? Differentiate between Love and Rayleigh seismic waves with the help of diagrams?	CO3
(b)	Explain the formation of seismic waves on the basis of elastic rebound theory with suitable diagrams?	
(c)	Correlate Richter scale and modified mercalli intensity scale? Define the magnitude of earthquake and explain how the magnitude can be measured in Richter scale based on observations?	
Q4	(20 marks)	

(a)	Define liquefaction. Discuss the effects and factors controlling the liquefaction? , Suggest methods for reducing liquefaction effects?	CO4
(b)	Consider a 4 storey RC office building located in Roorkee. The soil conditions are medium stiff and the entire building is supported on a raft. The R.C. frames are infilled with brick-masonry (SMRF). The lumped weight due to dead loads is 15 kN/m ² on floors and 12 kN/m ² on the roof. The floors are to cater for a live load of 3 kN/m ² on floors and 2.5 kN/m on the roof. Determine design seismic load on the structure as per new code. Plan area of the floors is 350 sq.m. and the height of each floor is 3.5m.	
(c)	How can we obtain response spectrum from the responses of spring mass oscillators having different natural frequencies embedded in the ground having same damping?	
Q5	(20 marks)	
(a)	How tuned mass damper can influence the response of tall structures during seismic movement?	CO5
(b)	Classify and explain the different base isolation systems used in structures?	
(c)	Enlist the ductile detailing consideration as per IS 13920: 1993 for design of flexural members?	